

GreenTech Solutions

Database Design & SQL Task

Total Time: 6 Hours

Total Marks: 20

Scenario

You work for a software development company that has secured a contract with **GreenTech Solutions**, a company specialising in sustainable living products and services.

GreenTech Solutions wants to develop a digital system to support:

- Users accessing the platform
- Information about eco-friendly products and practices
- Tracking sustainability activities
- Community engagement and interaction

You have been asked to design and document a **database solution** that could support this system.

Purpose of This Task

This task allows you to demonstrate that you can:

- Design a database from a real-world scenario
- Normalise data correctly
- Create clear database schemas
- Write and test SQL using Visual Studio Code and SQLite
- Explain your technical decisions clearly

You will submit **one evidence file** showing all stages of your work.

What You Must Submit

You must submit **one document** (Word or PDF) that includes **all sections below**, in order.

Your SQL must be written by you and tested before submission.

How You Will Work (Important)

You will use **Visual Studio Code (VSC)** with **SQLite** to create and demonstrate your database.

Software Setup

- Visual Studio Code
- SQLite extension by **Alex Covizzi**

Required SQLite Workflow

Before running any SQL, you must:

1. Create a database file, for example:

greentech.db

2. Create a SQL file, for example:

greentech.sql

3. Press **Ctrl + Shift + P**
4. Select **SQLite: Open Database**
5. Choose your **greentech.db** file

If you do not open the database first, your SQL will not run.

You should:

- Run SQL statements one at a time
- Check output before continuing
- Only include SQL in your evidence file once it works

Section 1: Unnormalised Data (0NF)

(Required – Not Marked)

Create **one unnormalised table (0NF)** that fits the GreenTech Solutions scenario.

Requirements:

- One table only
- At least **10 records**
- Clear duplication or repeating groups
- Fits the scenario clearly

This shows your starting point before improvement.

Section 2: Normalised Tables (3NF)

(Marked)

Using your 0NF data, redesign the database into **Third Normal Form (3NF)**.

You must:

- Create multiple related tables
- Remove duplicated data
- Use appropriate primary keys
- Use appropriate foreign keys
- Ensure tables are logically linked

You should include:

- Table names
- Example records (based on your original 0NF data)

Section 3: Database Schema

(Marked)

For **each table** you created in 3NF, provide a **database schema**.

Your schema must clearly show:

- Table name
- Column names
- Data types
- Primary keys
- Foreign keys

Schemas must be presented in a **clear table format**, similar to how schemas are shown in SQL Academy.

No SQL code is required in this section.

Section 4: SQL Implementation (DDL and DML)

(Marked)

You must now demonstrate SQL skills by writing and testing SQL based on your schema.

Your SQL must be relevant to your database design.

DDL – Database Structure

You must include examples of:

- **CREATE TABLE** (at least two tables)
- **ALTER TABLE** (add a column)
- **DROP TABLE** (one example, explained)

DML – Working with Data

You must include examples of:

- **INSERT**
- **SELECT** (with WHERE and ORDER BY)
- **JOIN**
- **UPDATE** (with WHERE)
- **DELETE** (with WHERE)

Your SQL must:

- Be correctly structured
- Use safe WHERE clauses
- Match your schema and scenario

Evidence Requirements

Your evidence file should include:

- Written explanations
- Tables and schemas
- SQL code
- Screenshots or copied output showing SQL ran successfully

Do not include broken or untested SQL.

How This Task Is Marked

Banded Mark Scheme (Out of 20)

This task uses **banded marking**.

Marks are awarded for **quality, accuracy, and clarity**, not just volume.

Band 1: Limited (1–5 marks)

- 3NF tables are unclear or incorrect
- Keys are missing or incorrect
- Schema is incomplete or confusing
- SQL shows limited understanding
- Explanations are weak or inaccurate

Band 2: Basic (6–10 marks)

- 3NF tables mostly correct with minor issues
- Primary keys present, foreign keys may be unclear
- Schema provided but lacks detail
- SQL includes basic DDL and DML
- Some correct use of WHERE and JOIN

Band 3: Secure (11–15 marks)

- Clear and correct 3NF design
- Keys used correctly
- Schema is clear and consistent
- Good range of DDL and DML
- SQL is mostly accurate and safe
- Explanations show understanding

Band 4: Strong / Excellent (16–20 marks)

- Fully correct and well-designed 3NF tables
- Professional-quality schema presentation
- Confident and accurate SQL
- Safe and precise UPDATE and DELETE
- Clear, well-written explanations using correct terminology

Final Student Checklist

Before submitting, check:

- I opened my SQLite database before running SQL
- My ONF table shows clear problems
- My 3NF tables remove duplication
- My schema is easy to read
- My SQL has been tested
- My UPDATE and DELETE statements are safe
- My explanations are clear